

Lecture 12

The 4 + 1 Model

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March 2, 2026

- The 4 + 1 model is composed of 5 distinct views.
 - Each view can be expressed with models reflecting their design.
- Below is a list of some acceptable models each view in the 4 + 1 model can use to express their design.
 1. The logical view:
 - Class diagrams.
 - Block diagrams.
 - Sequence diagrams.
 - Communication diagrams.
 2. The process view:
 - Activity diagrams.
 - Interaction diagrams.
 3. The development view:
 - Package diagrams.
 4. The physical view:
 - Deployment diagrams.
- ‘Patterns’ are reliable solutions for frequently encountered problems in design.
- ‘Architectural styles’ are equivalence classes describing systems which use similar patterns when solving a given problem.
- In use case diagrams:
 - ‘Use case controllers’ orchestrate what use case (or mode) the system is in.
 - * Use case controllers can be represented mathematically as Finite State Machines (FSMs)
- In class diagrams:
 - Boundary classes are interface hiding classes.